

A

Mini Project Report on

Design and Analysis of Rack and Pinion Steering System

By

Aditya S. Badgujar

T. Y. BTech Mechanical Engineering

(PRN: 1032181727; Div.: D; Roll No.: PD 52)

Guided by

Prof. Chetan K. Patil



Dr. Vishwanath Karad

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School of Mechanical Engineering

Dr. Vishwanath Karad

MIT World Peace University, Pune

2020 – 2021

Dr. Vishwanath Karad

MIT World Peace University, Pune

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CERTIFICATE

This is to certify that **Mr. Aditya S. Badgujar**, has successfully completed the mini project “**Design and Analysis of Rack and Pinion Steering System**” under my supervision, in the partial fulfilment of third year, ninth trimester - School of Mechanical Engineering of Dr. Vishwanath Karad MIT World Peace University, Pune for the academic year 2020-2021.

Date:12/06/2021

Place: Pune

Prof. Chetan K. Patil
Mini Project Guide

Prof. A.M. Elgandelwar
Examiner

Seal

Prof. (Dr.) S. T. Chavan
Head of School

Dr. Vishwanath Karad MIT World Peace University, Pune	Vision and Mission
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ACKNOWLEDGEMENT

It gives me great pleasure to present a mini project on “**Design and Analysis of Rack and Pinion Steering System**”. In presenting this mini project work, a number of hands helped me directly and indirectly. Therefore, it becomes my duty to express my gratitude towards them.

I am very much obliged to subject guide **Prof. Chetan K. Patil**, in School of Mechanical Engineering, for helping and giving proper guidance. His timely suggestions made it possible to complete this seminar for me. All efforts might have gone in vain without his valuable guidance.

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Aditya S. Badgujar

TY BTech Mechanical Engineering

PRN: 1032181727

Roll No: PD 52

CONTENTS

Sr. No.	Name of Sections	Page No.
	List of Figures	ii
	List of Tables	iii
	Nomenclature	iv
	Abbreviations	v
1.	Introduction	1
2.	Literature Survey	2
3.	3.1 Problem Statement	5
	3.2 Objectives	5
	3.3 Scope	5
4.	Methodology	6
5.	Material Selection	7
6.	Development And Execution	8
	6.1 Calculations for Helical Gearing Parameters	8
	6.2 Calculations for Gear Terminology	10
	6.3 Software Modelling & Analysis	11
	6.3.1 CAD Modelling	11
	6.3.2 Finite Element Analysis	13
7.	Results	17
8.	Conclusion	18
9.	References	19

LIST OF FIGURES

Fig. No.	Name of Figures	Page
1.	Manual Steering System Components	1
2.	Variable Rack	5
3.	Standard Rack	5
4.	Tooth Profile Sketch	11
5.	Pinion	11
6.	Center Rack	11
7.	Base Rack	12
8.	Final Assembly with Motion Links	12
9.	Safety Factor	13
10.	Von Mises Maximum Stress	14
11.	Bending Stress	14
12.	Reaction Forces	15
13.	Modal Analysis	15
14.	Contact Pressure	16

LIST OF TABLES

Table. No.	Name of Table	Page
1.	Material properties of Grey Cast Iron	7
2.	Material properties of Plain Carbon Steel	7
3.	Gear parameters for helical gears	8
4.	Gear Terminology	10
5.	Result Table	17

NOMENCLATURE

α	–	Transverse pressure angle for helical gear
ψ	–	Helix angle for helical gear
z_p	–	Number of teeth on helical pinion
m_n	–	Normal module of helical gear
b	–	Face width of gear tooth
σ_b	–	Bending stress
Y	–	Lewis's form factor
d_p	–	Pitch circle diameter
P_t	–	Tangential load
c_v	–	Velocity factor
c_s	–	Service factor
P_{eff}	–	Effective load
S_b	–	Strength in bending
m	–	Transverse module for helical gears
P	–	Circular Pitch
p	–	Diametral pitch
r	–	Root fillet radius
A	–	Addendum
D	–	Dedendum
t	–	Tooth thickness
S_{ut}	–	Ultimate tensile strength of material
S_{yt}	–	Yield strength of material

ABBREVIATIONS

FOS	–	Factor of Safety
IJSART	–	International Journal for Science for Science and Advance Research in Technology
IJEIT	–	International Journal of Engineering and Innovative Technology
RPM	–	Revolution per minute
CAD	–	Computer Aided Design
FEA	–	Finite Element Analysis

INTRODUCTION

Steering system is one of the key components of driving system that helps in maneuvering the direction of any vehicle effortlessly. Its function is to convert the rotating motion from the driver's side to translational motion with the help of rack and pinion assembly. These two components absorb the shocks from the tyre, and prevent further transmission of jerks to the steering wheel. Classification of steering system is based on two mechanisms, Davis Steering Mechanism and Ackerman Steering Mechanism

Traditionally, Davis Steering Mechanism was under consideration but, as it has sliding joints the steering gears wear out easily. Ackerman Steering Mechanism have only turning pairs and do not require space while in operation, therefore they are more in conventional use.

The parts involved in steering system are listed below,

1. Pinion
2. Rack
3. Tie Rod
4. Mounting brackets

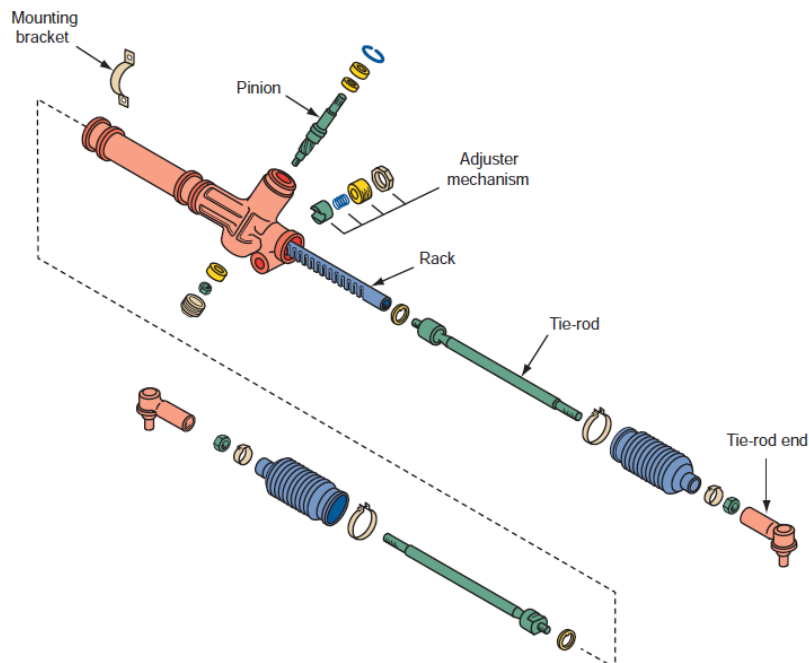


Fig. 1: Manual Steering System Components [2]

LITERATURE SURVEY

1) Investigation of Load Carrying Capacity for Steering System with Polymer Helical Rack and Pinion Gear [3]

Authors- Yogesh Ashok Ingale, A.P. Kadam, Digvijay G. Bhosale and A.P. Bhosale

Journal name: Springer International Publishing AG

Published Year: 2018

Page no.: 891 to 902

Content:

In this research paper the team has investigated on steering gearbox (consisting of a rack and pinion gear unit) of a hatchback car for the increase in transmissibility power level.

Comparative study of physical properties of Nylon 6/6 and DELRIN gear materials is performed on the gears by using approaches such as FEA and experimental analysis.

Separate calculations are performed for both the materials and the finite element analysis is done in ANSYS software.

Boundary Conditional Forces, Von-Mises stress & tooth Deformation are measured analytically and through software for respective materials. Strain results and weight and cost results along with the lifespan of gears are compared.

Conclusion:

Finally, the paper concludes, that the polymer gears are weight and cost effective than the cast iron or steel gears. The value of factor of safety for the both the materials is also observed more than recommended for bending load.

2) Design and Analysis of Steering Gear and Intermediate Shaft for Manual Rack and Pinion Steering System [6]

Authors- Thin Zar, Thein Hlaing, HtayHtay Win, Myint Thein

Journal name: International Journal of Scientific and Research Publications, Vol. 7, Issue 12

Published Year: 2017

Page no.: 861 to 882

Content:

This paper investigates the characteristics of a rack and pinion gear system mainly focused on bending and contact stresses of the pinion gear and rack bending stress using analytical and finite element analysis methods.

The 3D solid models for different materials are generated by SolidWorks software and the numerical solution is done by ANSYS where the analytical investigation is based on Lewis stress formula.

Both the results are compared using AGMA (American Gear Manufacturers Association) equation.

Fatigue analysis of intermediate steering shaft is done to find the life of the intermediate steering shaft in cycles and determined the factor of safety of the shaft so as to increase the performance and efficiency of gears.

Due to the vehicular weight the rack has possibility of bending so Von-Mises Stresses and Deformation for rack are performed. Gear tooth analysis is done so as to test the strength in teeth of gears.

All of the above-mentioned analysis are performed with three different materials

- 1) AISI 4340 (Stainless steel)
- 2) Aluminium Alloy
- 3) Gray Cast Iron

Conclusion:

From the structural contours of ANSYS, it can be observed that the maximum strains are also occurred at the meshing area of the rack and pinion gear. The von-Mises stresses on gear pair using different materials are nearly the same. The analysis shows that the strain has minimum with AISI 4340 steel. Therefore, AISI 4340 steel is suitable material in this designed steering gear pair.

3) Design and Simulation of 4 Wheel Steering System

[5]

Authors- Saket Bhishikar, Vatsal Gudhka, Neel Dalal, Paarth Mehta, Sunil Bhil, A.C. Mehta

Journal name: International Journal of Engineering and Innovative Technology (IJEIT)

Volume 3, Issue 12

Published Year: 2014

Page no.: 351 to 367

Content:

The authors have developed an innovative 4-wheel steering design to implement a mechanism that can serve the purpose of changing in-phase and counter-phase steering of rear wheels depending upon the conditions of turning and lane changing with respect to front wheels, thus enhancing the maneuverability of HONDA CIVIC sedan car in accordance with its speed.

A double rack and pinion system for rear wheels enclosed within a casing, connected to the steering column by a combination of a bevel gear assembly and telescopic shaft is designed.

The position of wheels at low speeds and high speeds are compared and based on that the in phase and counter phase steering are designated with the help of three bevel gears assisted to front rack and pinion mechanism connected to front pinion, steering rod and rear pinion.

Rear wheel system consists of two racks with two pinions. One of the racks will be in front of the rear wheel axis (primary rack) and the other will be behind the axis (secondary rack).

When car is running is speed below 35kmph the hydraulic actuator will keep the pinion on the secondary rack to keep both wheels out of phase. As soon as the speed of car exceeds 35kmph, hydraulic actuator shifts the pinion to primary rack and this done easily with the help of Telescopic Shaft.

Considering the standard specifications of HONDA CIVIC all the calculations and analysis are done.

Further bevel gear calculations are done along with Two major analyses and cost analysis

1) Deformation analysis

2) Stress analysis

Conclusion:

This system assists in high-speed lane changing and better cornering eliminating the problems faced in sharp turning. It reduces the turning circle radius of the car and gives better maneuverability and control while driving at high speeds, thus attaining neutral steering.

4) Research and Structural optimization method of rack and pinion steering gear with variable ratio. [4]

Authors- Ziru Niu, Gangyan , Jian Hu, Tianli Yan, Siyu Zhang, and Jiangjiang Tan

Journal name: Applied Mechanics and Materials Vol. 664

Published Year: 2014

Page no.: 98 to 103

Content:

In this paper author has focused on variable gear ratio steering transmission. The Structural optimization method is used for the rack and pinion steering gear with variable ratio. The components focused in this are pinion gear variable gear ratio rack and tie rod. The modelling operation is done in CATIA while the analysis is performed in ANSYS. It is based on Boolean subtraction operation based on Enveloping Principle Simulation Method.

Finally, the situation of deformation and stress-strain have been analysed, during the meshing process of variable ratio gear and rack by conducting static analysis of the whole model of steering in workbench. The structure of the tie rod can be improved, and a kind of cheaper material whose strength is lower than gear and rack can be used to save costs.

Conclusion:

Mainly in this paper stress strain properties are considered for further improvement, also the author suggests to go for cheaper material for the gears and maintaining the quality for material in the tie rods.

3.1 Problem statement

A single gear ratio rack requires a greater number of turns in order to swivel the tyres. So, to eliminate this problem, implementation of variable gear ratio rack is necessary, having a maximum torque of 35kNm and 12rpm input speed from steering wheel.

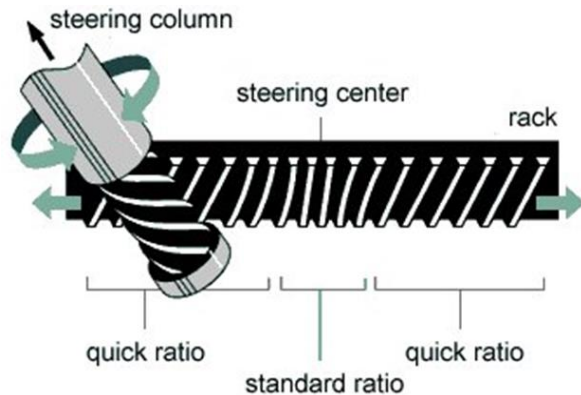


Fig. 2: Variable Rack

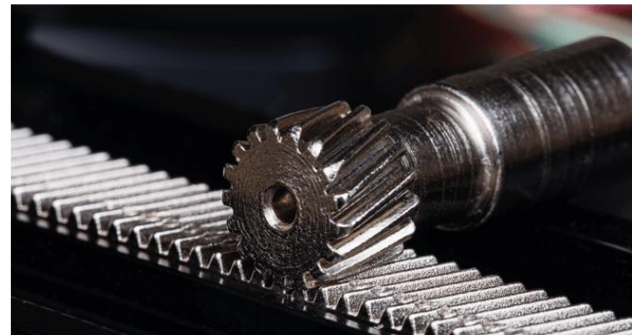


Fig. 3.: Standard Rack

3.2 Objectives

1. To design and model a variable steering system.
2. To find the maximum stress and deflection on the rack due to vehicle load and forces generated by the vehicle analytically and by simulation.
3. To analyze modal properties of the generated model.

3.3 Scope

This system can be implemented, in all types front wheel steering drive vehicles where minimal amount of steering force is required.

Further improvement to this project can be done on the intermediate shaft. An epicyclic gear train can be introduced on it so as to prevent the locking of gears in power steering system.

METHODOLOGY

In this mini project, the main components for analysis are the pinion, center rack and base rack. Therefore, suitable material is selected for these components.

Boundary conditions are analyzed and proper loads are applied and their values are taken under consideration for analytical as well as software solution. As in this model gears are involved the required gearing parameters are calculated for further force and stress analysis.

Modelling of the components is done in Autodesk Fusion 360 CAD Design Workspace and further finite element analysis is performed in Autodesk Fusion 360 Simulation Workspace.

The static stress analysis along with modal frequency analysis is performed for the steering system.

At the end, the analytical and software solutions are compared with each other and the results are concluded.

MATERIAL SELECTION

Material selected for gears = Grey Cast Iron (FG 260) [8]

(Alloy of carbon and iron, Si, Mn, Ph, Sulphur)

Grey cast iron has high compressive strength, resistance to deformation, resistance to oxidation and has a high melting point of 1100°C to 1200°C, hence it is most suitable material for gears.

Table 1: Material properties of Grey Cast Iron

Parameter	Value
S_{ut}	700 N/mm ²
S_{yt}	420 N/mm ²
Min. tensile strength	260 N/mm ²
Poisson's ratio	0.26
Density	7.125 kg/mm ³
BHN	197

Material selected for shaft = Plain Carbon Steel (FE 620) [8]

Plain Carbon Steel has good ductility, high strength, wear resistance and has low cost, hence is best suited for manufacturing shafts.

Table 2: Material properties of Plain Carbon Steel

Parameter	Value
S_{ut}	620 N/mm ²
S_{yt}	380 N/mm ²
FOS	2.5

DEVELOPMENT AND EXECUTION

6.1 Calculations for Helical Gearing Parameters

Table 3: Gear parameters for helical gears

Sr. No.	Parameter	Symbol	Value
1	Pressure angle	α	20°
2	Helix angle	ψ	18°
3	Pinion Teeth	z_p	20
4	Factor of Safety	F.O.S.	1.5

Input Speed (rpm) for a steering system ranges from 15 to 8 rpm, therefore on an average it is considered as 12 rpm for this steering system

Torque is considered from the force applied by the driver on the steering wheel, where the average human force that is applied on a steering wheel is taken as 200N.

Face width of gear lies between $8 \times m_n$ to $12 \times m_n$ [1]

$\therefore$ Considering face width as $b = (11 \times m_n)\text{mm}$

Torque = (Force on Steering $\times$ Radius of steering) = $200\text{N} \times 175\text{mm} = 35000 \text{ Nmm}$

Bending stress = $\sigma_b = \frac{S_{ut}}{3} = \frac{700}{3} = 233.334\text{N/mm}^2$...Equation (1)

Lewis Factor = $Y = 0.320$ for 20 teeth [1]

Pitch circle diameter = $d_p = z_p \times \frac{m_n}{\cos(\psi)} = 20 \times \frac{m_n}{\cos(18)} = (21.0292 \times m_n) \text{ mm}$...Equation (2)

Tangential Load = $P_t = 2 \times \frac{M_t}{d_p} = 2 \times \frac{35000}{21.0292 \times m_n} = \left(\frac{3328.6978}{m_n} \right) \text{N}$...Equation (3)

Velocity Factor = $c_v = \frac{3}{3+v}$ [1]

Assuming trial velocity as 3m/s

$$\therefore \text{Velocity Factor} = c_v = \frac{3}{3 + v} = \frac{3}{3 + 3} = 0.5$$

$$\text{Service Factor} = c_s = 1.25 \quad [1]$$

$$\text{Effective Load} = P_{\text{eff}} = \frac{c_s}{c_v} \times P_t = \frac{1.25}{0.5} \times \left(\frac{3328.6978}{m_n} \right) = \left(\frac{8321.7445}{m_n} \right) \text{N} \quad \dots \text{Equation (4)}$$

$\therefore$ Strength in bending is given as,

$$\text{Bending Strength} = S_b = m_n \times b \times \sigma_b \times Y_p = m_n \times (11 \times m_n) \times 233.334 \times 0.320$$

$$\therefore S_b = (821.3356 \times m_n^2) \text{N} \quad \dots \text{Equation (5)}$$

Also, strength in bending is given by,

$$S_b = P_{\text{eff}} \times \text{F. O. S.}$$

$\therefore$ From equations (4) and (5),

$$(821.3356 \times m_n^2) = \left(\frac{8321.7445}{m_n} \right) \times 1.5 \quad \text{from table}$$

$$\therefore \text{Normal Gear Module} = m_n = 2.477 \approx 2.5 \quad (\text{Section 18.4 Tooth Proportions V.B. Bhandari})$$

$$\therefore \text{Pitch circle diameter} = d_p = (21.0292 \times 2.5) = 52.573 \text{ mm} \quad \dots \text{From equation (2)}$$

$$\text{Tangential Load} = P_t = 2 \times \frac{M_t}{d_p} = 2 \times \frac{35000 \text{ Nmm}}{52.573 \text{ mm}} = 1331.4819 \text{ N} \quad \dots \text{From equation (3)}$$

$$\text{Effective Load} = \frac{c_s}{c_v} \times P_t = \frac{1.25}{0.5} \times 1331.4819 \text{ N} = 3328.6978 \text{ N} \quad \dots \text{From equation (4)}$$

$$\text{Bending Strength} = (821.3356 \times 2.5^2) = 5133.3475 \text{ N} \quad \dots \text{From equation (5)}$$

6.2 Calculations for Gear Terminology

Table 4: Gear Terminology

Sr no	Gear Terminology	Symbol	Relation with Module	Value
1	Transverse module	m	$\frac{m_n}{\cos(18)}$	2.6286 mm
2	Normal module	m_n	—	
3	Circular Pitch	P	$\pi \times m$	8.2581 mm
4	Diametral Pitch	p	$\frac{1}{m}$	0.3804 mm
5	Root Fillet Radius	r	$0.38 \times m$	0.95
6	Backlash	b	—	0.1
7	Addendum	A	$1 \times m_n$	2.5 mm
8	Dedendum	D	$1.25 \times m_n$	3.125 mm
9	Tooth thickness	t	$\frac{\pi}{2} \times m_n$	3.927 mm

6.3 Software Modelling & Analysis

6.3.1 CAD Modelling

The model is designed and modelled in Autodesk Fusion 360 CAD Workspace. [9]

The parts that this design consist are,

- 1) Pinion
- 2) Center Rack
- 3) Base Rack

Whole assembly is shown below:

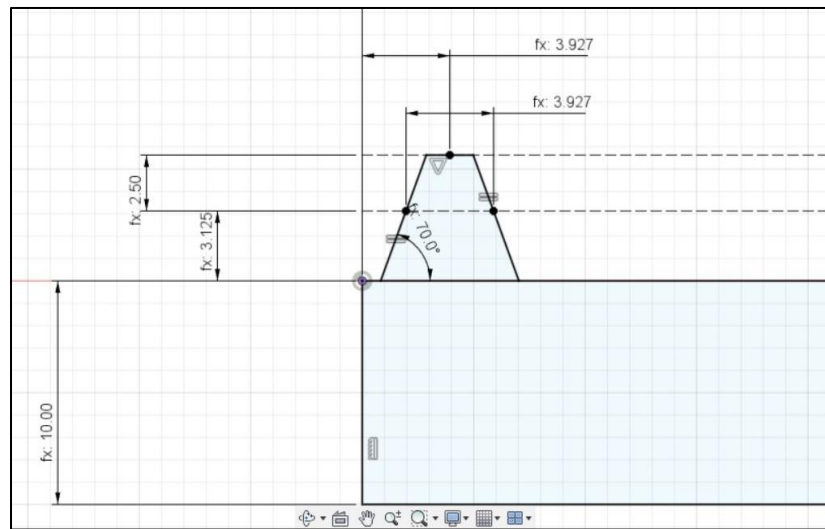


Fig. 4: Tooth Profile Sketch

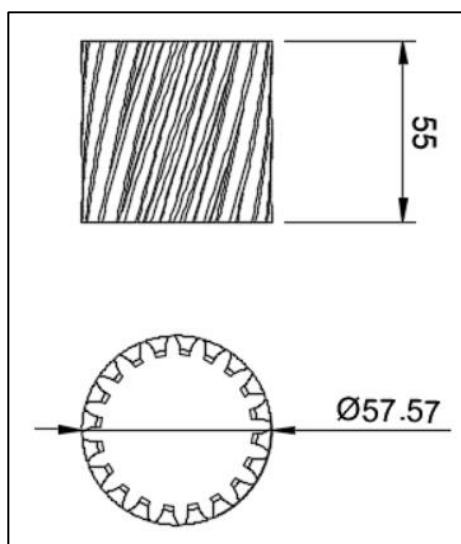


Fig. 5: Pinion

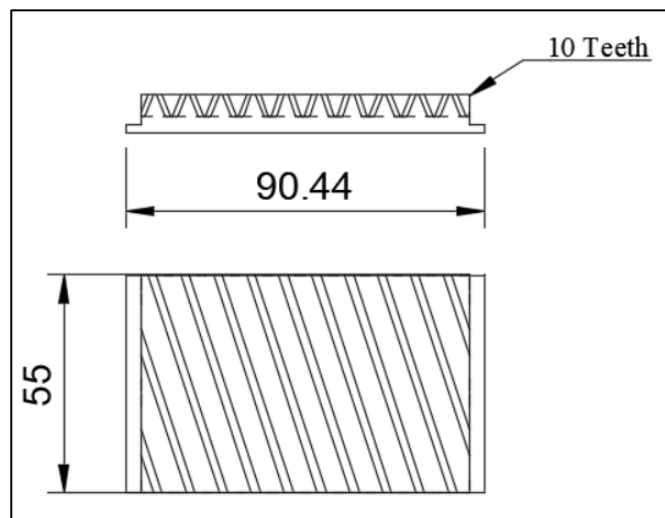


Fig. 6: Center Rack

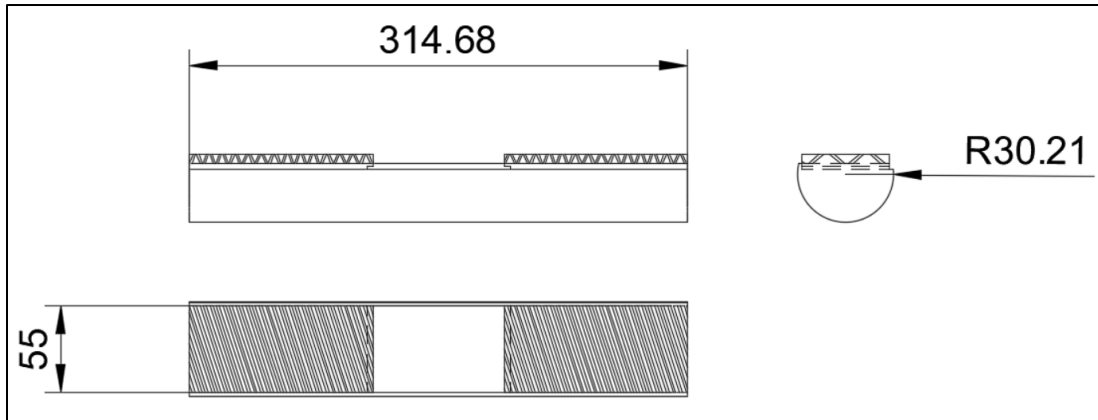


Fig. 7: Base Rack

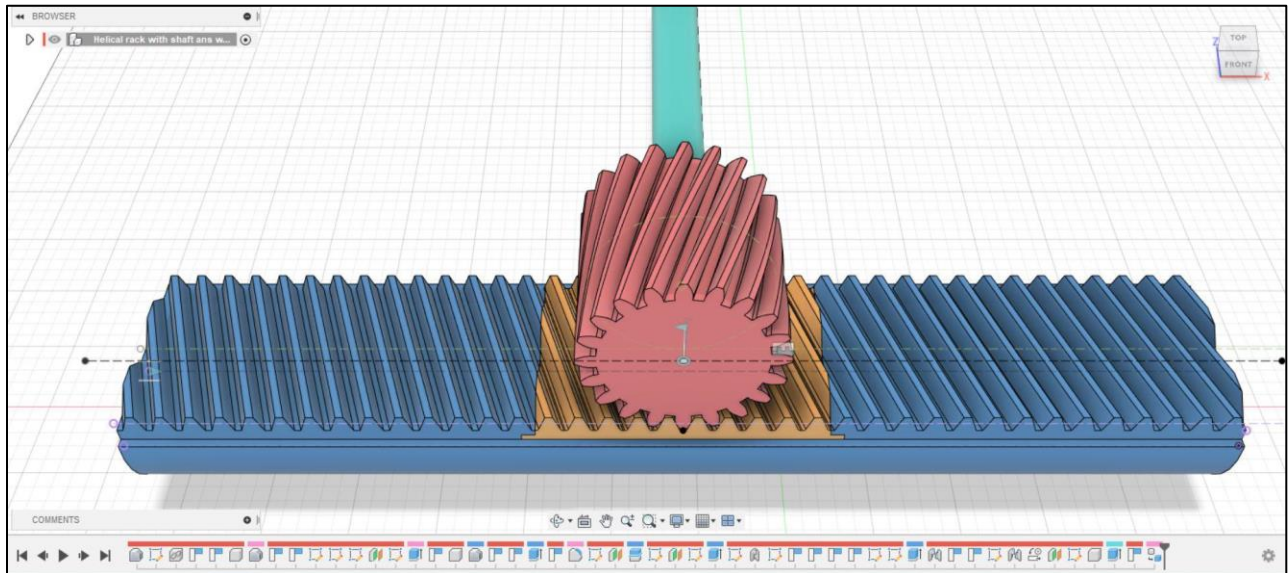


Fig. 8: Final Assembly with Motion Links

6.3.2 Finite Element Analysis

The finite element analysis is performed in Autodesk Fusion 360 Simulation Workspace. [9]

Analysis is done on a particular component to find out the necessary parameters of the model by applying proper boundary conditions such as, external loads, torque, thermal constants, etc.

The analyses carried out in this project are,

1. Safety Factor
2. Von Mises Maximum Stress
3. Maximum Bending Stress
4. Reaction Forces
5. Modal Analysis
6. Contact Pressure

1) Safety Factor

The safety factor defines, by how much a particular component can sustain a certain load. In other words, it is a factor by which, the applied stress on the component is less than the yield strength of the material selected for the component.

For this model the safety came out to be around 1.328 which denotes that is a safe design and can sustain higher torques.

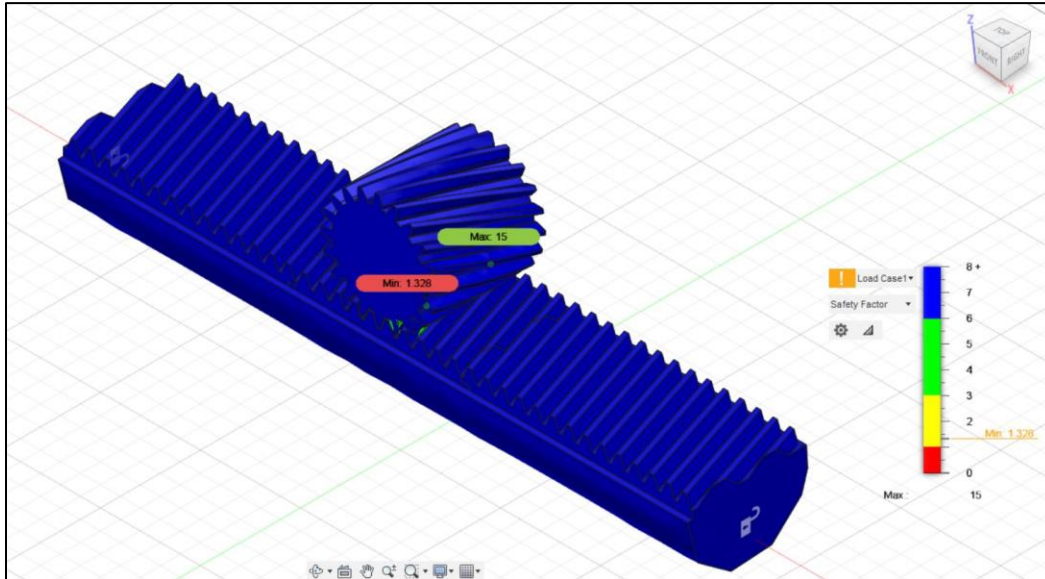


Fig. 9: Safety Factor

2) Von Mises Stress

Von Mises Stress is a criterion, which tells that a component will yield the stress or will undergo fracture. So, in this mini project, as the maximum stress on the rack is obtained as 324.4 MPa and taking factor of safety under consideration, the rack on which the stress is acting will yield the stress.

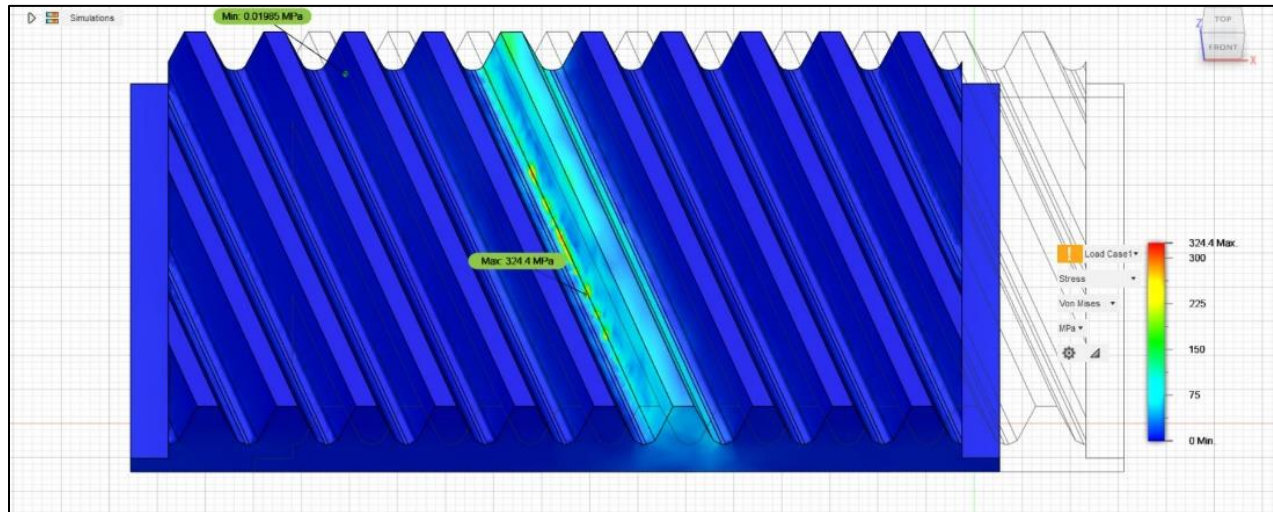


Fig. 10: Von Mises Maximum Stress

3) Bending Stress

When an external moment or force acts on a component, it tends to bend and deform. Thus, to avoid this, bending stress is calculated analytically and later compared to the software results. The software results obtained for this rack is 232.3MPa.

The bending stress analytically is given by,

$$\text{Bending stress} = \sigma_b = \frac{S_{ut}}{3} = \frac{700}{3} = 233.334 \text{ N/mm}^2 = 233.334 \text{ MPa}$$

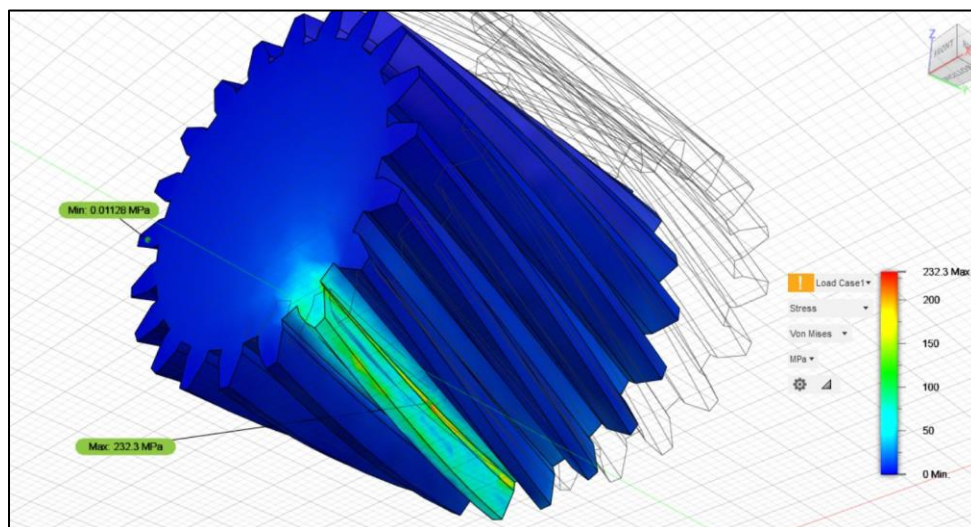


Fig. 11: Bending Stress

4) Reaction Forces

Reaction force can be defined as, the counter force so as to balance the external load or forces acting on a part. It is related to Newton's third law of motion that is 'Every action has an equal and opposite reaction'.

In this mini project, the reaction forces obtained are on the base rack which can be equated to the forces that come from the frictional resistance from the wheels on the road.

The maximum reaction force is 4.714N and can be seen in the figure given below.

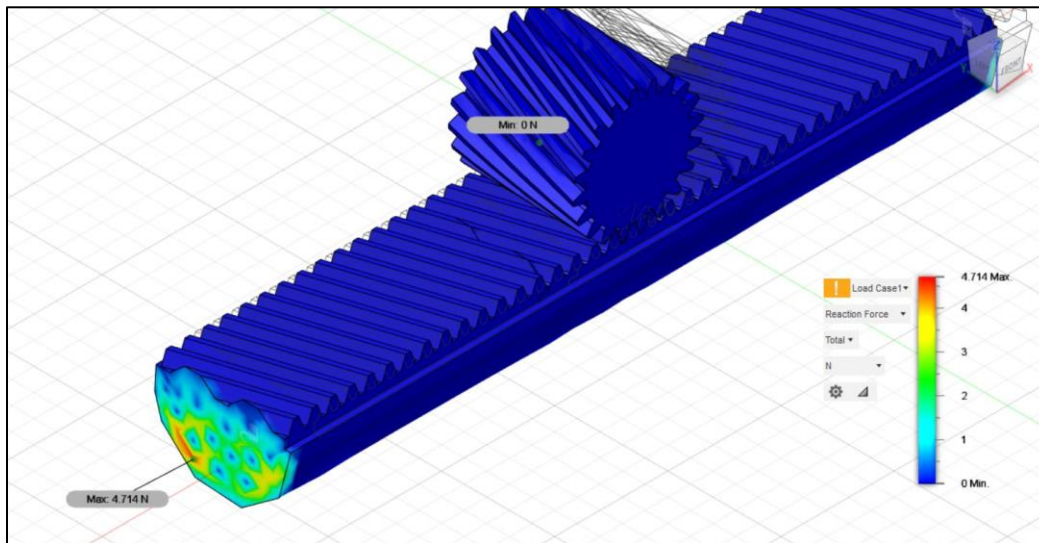


Fig. 12: Reaction Forces

5) Modal Analysis

Modal analysis helps to determine the vibrational characteristics of components. It depicts the movement of different parts of structure under loading (dynamic loading) condition. Below given are the 8 modes and their corresponding frequencies obtained for this rack and pinion assembly.

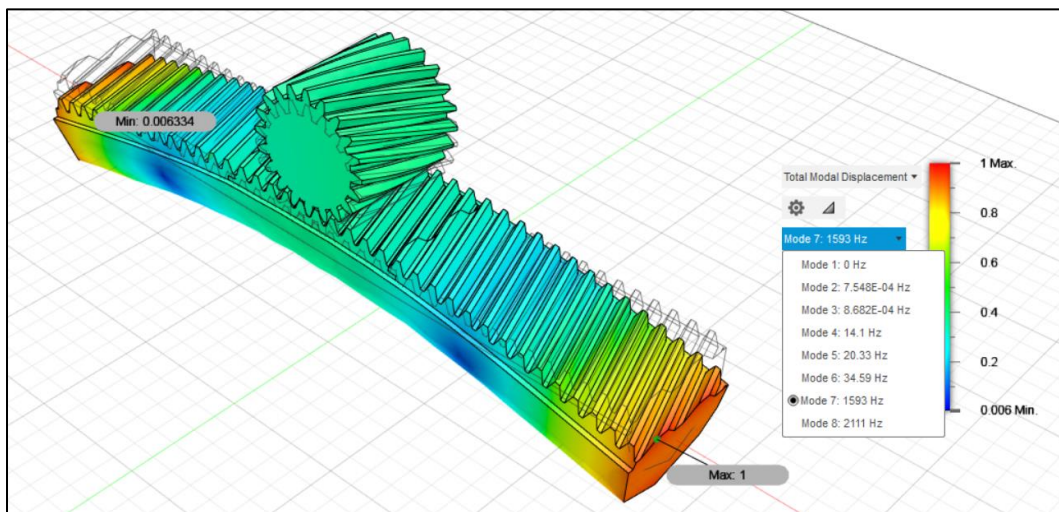


Fig. 13: Modal Analysis

6) Contact pressure

The pressure generated during contact between two parts is called as the contact pressure. In this project there is contact between the pinion and center rack, and the maximum value obtained is 404.5MPa.

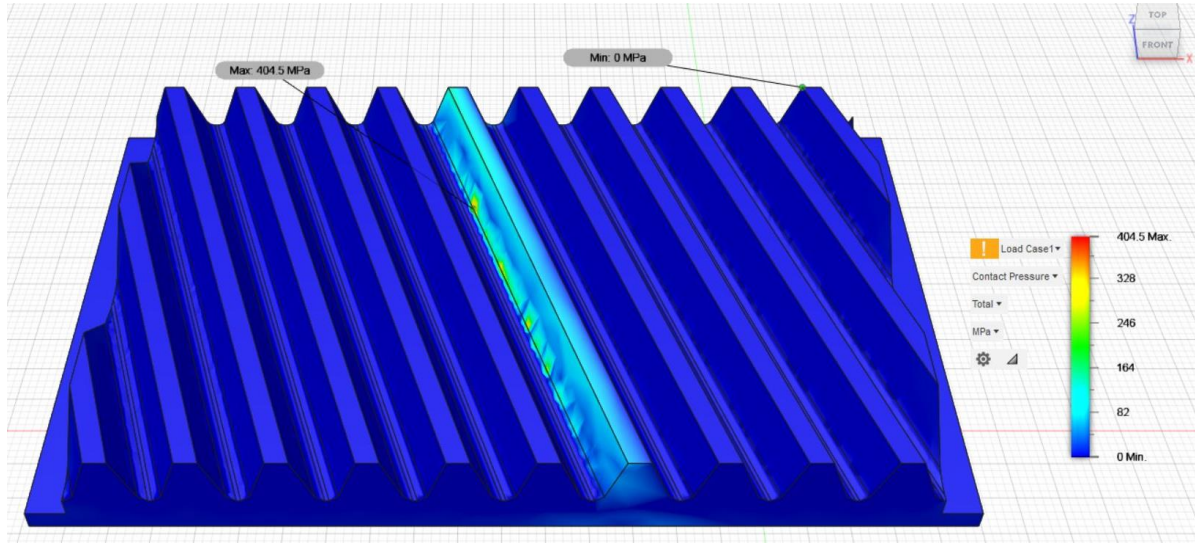


Fig. 14: Contact Pressure

Results

The Desired results that are required for a safe design of rack and pinion steering system are obtained analytically and through software.

Below is the tabulated representation and comparison between both the static analysis results.

Table 5: Result Table

Parameters	Analytical results	Software Results	Comment
Factor of Safety	1.5	1.328	-
Strength	$S_{yt} = 420 \text{ N/mm}^2$	$Stress_{max} = 324.4 \text{ N/mm}^2$	As $Stress_{max} < S_{yt}$ the design satisfies the strength properties
Bending Strength	$\sigma_b = 233.334 \text{ N/mm}^2$	$\sigma_b = 232.334 \text{ N/mm}^2$	Result matches the attained theoretical value
Contact Pressure	-	404.5 N/mm^2	Maximum contact pressure obtained on the flank of the variable rack.

Conclusion

Stress analysis including Von Mises and Bending stress is performed successfully on the steering system. The factor of safety that was considered for analytical solution nearly matches the software results. The maximum stress obtained in software analysis is less than the yield strength, this states that the designed model is safe in terms of strength and grey cast iron is a suitable material for gear manufacturing.

The factor of safety can be modified to 2.5 to 3 so as to attain better results and strength.

The modal analysis performed on the system determined the exact natural frequencies for the steering system assembly in various mode shapes.

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- **Web Links**

- [9] Autodesk Fusion 360 Support: <https://www.autodesk.com/products/fusion-360/resources>